Sahaj Bhadja

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Summary

Junior software engineer with STEM degrees and 2+ years developing applications in collaborative Agile teams. Strong **C++** fundamentals including **OOP**, **STL**, **memory management**, and **design patterns**, with experience in modeling/simulation, debuggers, and CI/CD testing infrastructure.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Data Structures, Object-Oriented Programming, Operating Systems, Computer Networks

Experience

Aasma Technology Solutions — **India**
C++ Software Engineer - (SecureSync) — *August 2023 – July 2024*

- Designed and implemented software modules using **OOP design patterns** (Singleton, Factory) and algorithmic optimization, improving API response by **30%** while working in **Agile Scrum** with cross-functional teammates on complex integration problems.
- Maintained **20+ unit and integration test suites** with **PyTest**, Mockito, and **Jenkins CI/CD** across Windows and Linux, using debuggers and test infrastructure to troubleshoot failures and increase release confidence by **68%**.

USC AD Labs — **Los Angeles, CA**
Software Engineering Assistant — *January 2025 – May 2025*

- Built **modeling and simulation** evaluation pipelines across **100K+ records**, applying data structures and algorithms to validate behavioral outcomes, reproduce test scenarios, and analyze results in collaborative research engineering environments.
- Contributed technical analysis and documentation while adapting solutions to evolving requirements, critically thinking through software issues and communicating findings clearly to multidisciplinary engineering partners.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed and enhanced production application code handling **10K+ daily requests**, implementing **multithreaded** batch processing with fault-tolerant execution and utilizing testing workflows to debug peak-load issues in Agile delivery cycles.
- Collaborated on requirements analysis and source-code improvements for existing application flows, reducing manual turnaround by **60%** through iterative development, code reviews, and structured troubleshooting in a team environment.

Pandit Deendayal Energy University — **India**
Software Engineer - Research Fellowship — *January 2023 – May 2024*

- Developed and benchmarked novel **consensus algorithms** in simulation environments, applying **C++**-style systems thinking, data structures, and performance analysis to reduce risk exposure by **25%** in benchmark workloads.
- Contributed to peer-reviewed systems research presented at IISc Bengaluru, solving routine and moderately complex software problems while referring deeper architecture questions to senior collaborators.

MedDevice Test Systems — **India**
Software Engineer Intern — *January 2023 – April 2023*

- Implemented **C/Python** test libraries and simulators for device communication scenarios, encoding lifecycle properties with deterministic logs and validation checks while integrating automated suites into **Git**-based CI pipelines.
- Used debuggers and structured test infrastructure to troubleshoot intermittent failures, creating documentation and coverage reports that improved reliability across regulated production-support software workflows.

Skills

C++ & Software Design: **C++**, C, OOP, design patterns, STL, data structures, algorithms, memory management, pointers, templates (familiar), debugging, gdb (familiar), problem solving, critical thinking
Development Practices: Agile, Scrum, **Git**, Jira (familiar), Bamboo (familiar), Jenkins, CI/CD, unit testing, integration testing, PyTest, Mockito, source code development, large codebases (familiar), modeling & simulation (familiar), AFSIM (familiar)
Systems: Linux, Windows, Bash, Java, Python, multithreading, concurrency, cross-platform development, technical documentation, collaborative teamwork, requirements analysis

Projects

CPU Scheduling Simulator Live | *C++, OOP, Modeling & Simulation*

- Implemented preemptive scheduling algorithms in **C++** with standard-library data structures, interactive simulation controls, and correctness testing across **1K+** scenarios, reducing average turnaround time by **25%** through performance analysis.

SimTrace Engine | *C++, Concurrency, Debug Infrastructure*

- Built a **C++** event-simulation harness with queue-depth metrics and timing analysis, using debuggers and test infrastructure to troubleshoot bottlenecks and validate correctness under bursty concurrent workloads in complex code paths.

Chat2SQL Code | *Java, Large Codebase, Docker, Git*

- Worked across a multi-module **Java** codebase with **Git**, Maven, Docker, and automated validation, navigating complex application structure to extend database tooling and improve developer workflows on Windows, macOS, and Linux.